Camino: secuencia $(v_0, e_1, v_1, \dots, e_p, v_p)$ tq

nodos arcos

$e_k \in \{v_{k-1}v_k,\ v_k v_{k-1}\}$

Si uv ⇒

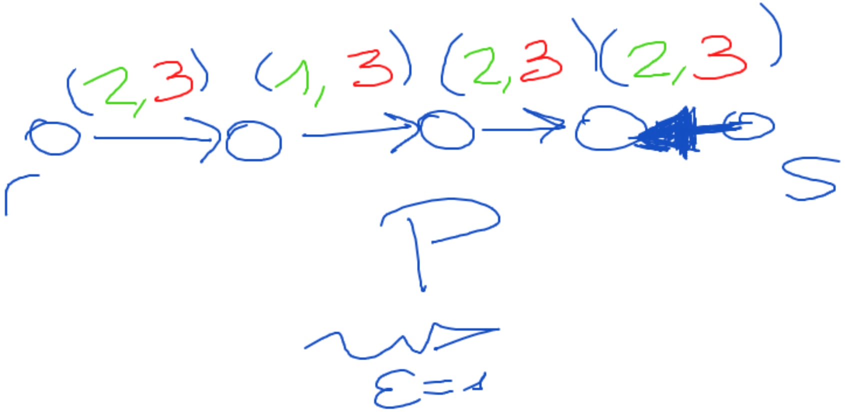
Camino aumento de r a s

$$\begin{cases} x_{uv} < c_{uv} & : uv \text{ directo} \\ x_{uv} > 0 & : uv \text{ reverso} \end{cases}$$

$\varepsilon_1 := \min\{c_{uv} - x_{uv} : uv \text{ directo}\}$

$\varepsilon_2 := \min\{x_{uv} : uv \text{ reverso}\}$

$\varepsilon(P) := \min\{\varepsilon_1, \varepsilon_2\}$

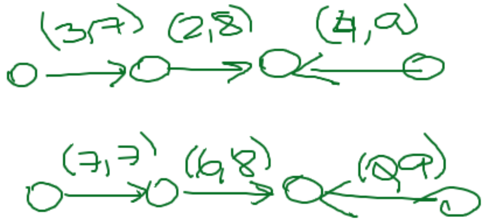


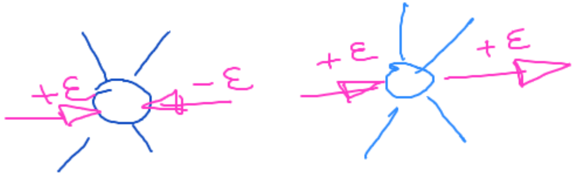
Propiedad Si P es un camino rs-aumentador entonces existe un flujo de valor $f(x)+\varepsilon(P)$

Prueba Sea

$$x'(uv) = \begin{cases} x(uv) & : \ uv \notin P \\ x(uv)+\varepsilon(P) & : \ uv \in P,\ uv \text{ es directo} \\ x(uv)-\varepsilon(P) & : \ uv \in P,\ uv \text{ es reverso} \end{cases}$$

$\varepsilon = 4$





Probaremos que x' es rs-flujo viable

- el flujo es no negativo en cada arco
- se respetan las capacidades
- $x^{+}_{-}(u) = 0 \quad \forall\, u \notin \{r,s\}$

∘ uv directo :

$$x'(uv) = x(uv) + \varepsilon(P)$$
$$\leq x(uv) + \varepsilon_1$$
$$= x(uv) + \min\{c_{uv} - x_{uv} : uv \text{ directo}\}$$
$$\leq \cancel{x(uv} + c_{uv} - \cancel{x_{uv}}$$
$$\leq c_{uv}$$

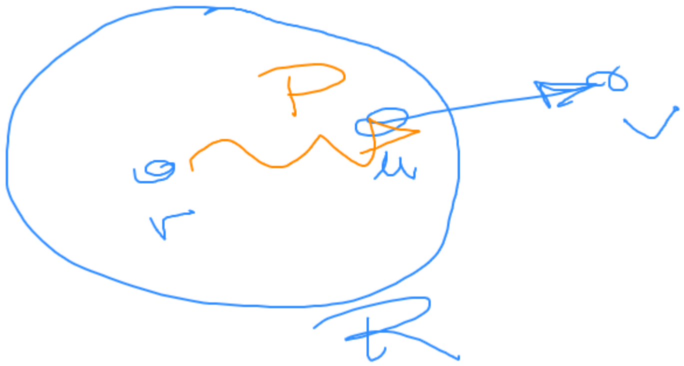
∘ uv reverso

$$x'(uv) = x(uv) - \varepsilon(P)$$
$$\geq x(uv) - \min\{x_{uv} : uv \text{ reverso}\}$$
$$\geq x_{uv} - x_{uv}$$
$$= 0$$

Teorema (MFMC) Existe flujo rs-viable x, tq $f(x) = c(R)$.
rs-corte $\partial^-(R)$

Prueba Sea x un rs-flujo máximo. Sea

$$R = \{ u : \text{existe camino } ru\text{-aumentador} \}$$

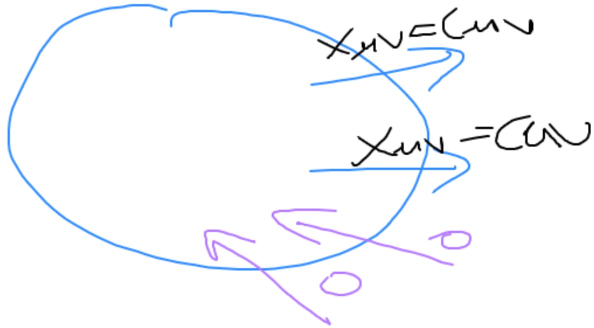


Note que $r \in R$, $s \notin R$. Probaremos que $x_{uv} = c_{uv}$ para todo $uv \in \partial^-(R)$. Suponga por contradicción que existe $uv \in \partial^-(R)$ tq $x_{uv} \neq c_{uv}$. Como $u \in R$, existe camino ru-aumentador P. Entonces $P \circ uv$ es un camino rv-aumentador, luego $v \in R$, contradicción.

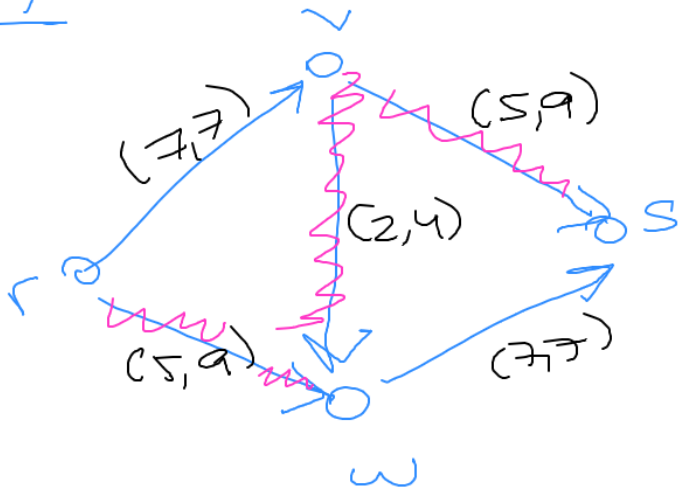
Análogamente, $x_{uv} = 0$ para cada $uv \in \partial^+(R)$.

Luego,

$$
\begin{aligned}
f(x) &= x^-_+(R) \\
&= x^-(R) - x^+(R) \\
&= c(R) - 0 \\
&= c(R) \mathbin{/\!/}
\end{aligned}
$$

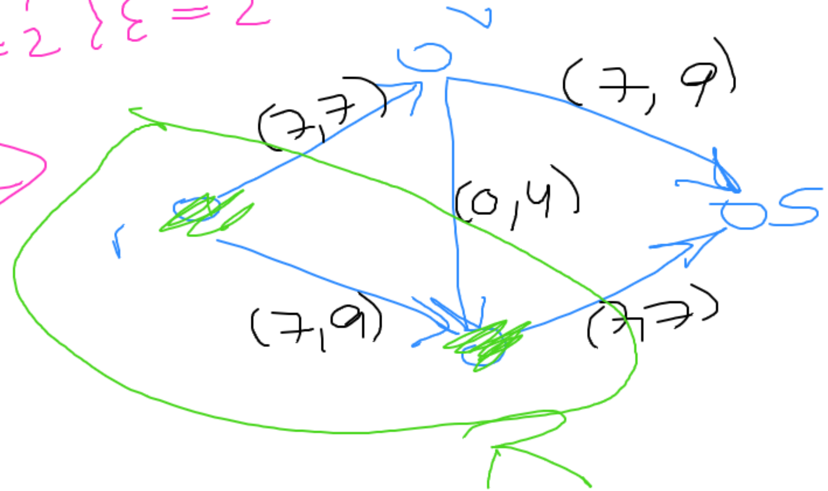


Ejm



$\left.\begin{array}{l}\varepsilon_1 = 4 \\ \varepsilon_2 = 2\end{array}\right\} \varepsilon = 2$

$\Longrightarrow$



Obs Si $c : E(D) \to \mathbb{Z}^+$ entonces existe un flujo maximo entero.

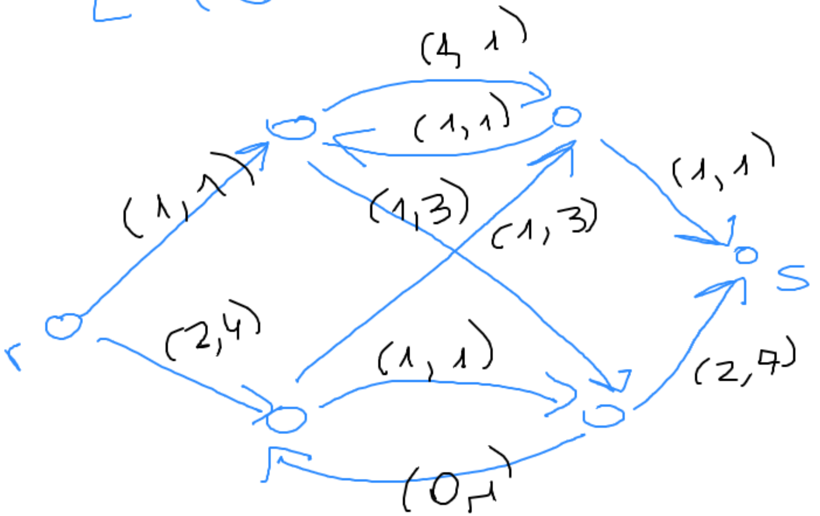
Algoritmo de Ford & Fulkerson

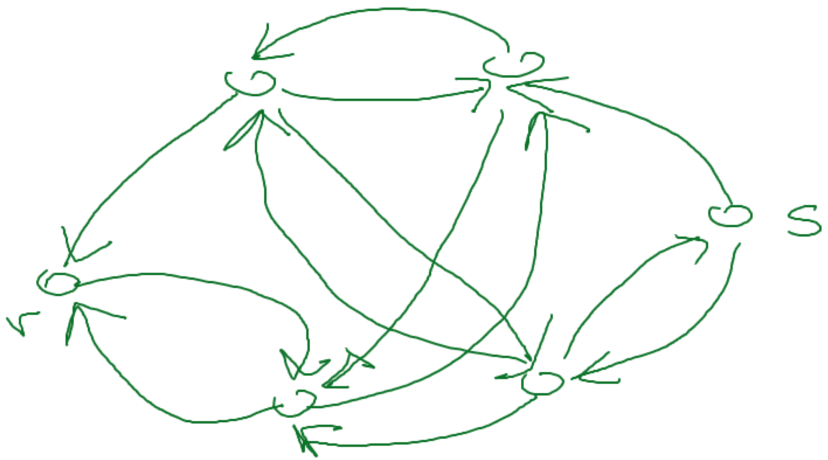
x é fluxo viável de grafo G, definimos $G(x)$

$V(G(x)) = V(G)$

$E(G(x)) = \{uv \in E(G) : x_{uv} < c_{uv}\} \cup \{vu : uv \in E(G) : x_{uv} > 0\}.$

Ex





caminhos aumentadores $\Longleftrightarrow$ caminhos de r a s.

$G \longrightarrow G(x)$

CaminodeAumento (G, c, x, r, s)

Construir $G(x)$

$P \leftarrow$ Encontrar Camino $(G(x), r, s)$

devolver P

EnviaFlujo (G, c, x, r, s, P)

Para todo $uv \in P$

Si uv es directo

$x_{uv} += \varepsilon(P)$

Caso contrario

$x_{vu} -= \varepsilon(P)$

devolver x

FordFulkerson (G, c, r, s)

$x \leftarrow 0$

Repetir

$P \leftarrow$ CaminodeAumento(G, c, x, r, s)

Si $P = \emptyset$ devolver x

$x \leftarrow$ EnviaFlujo(G, c, x, r, s, P)

si es entero

entero

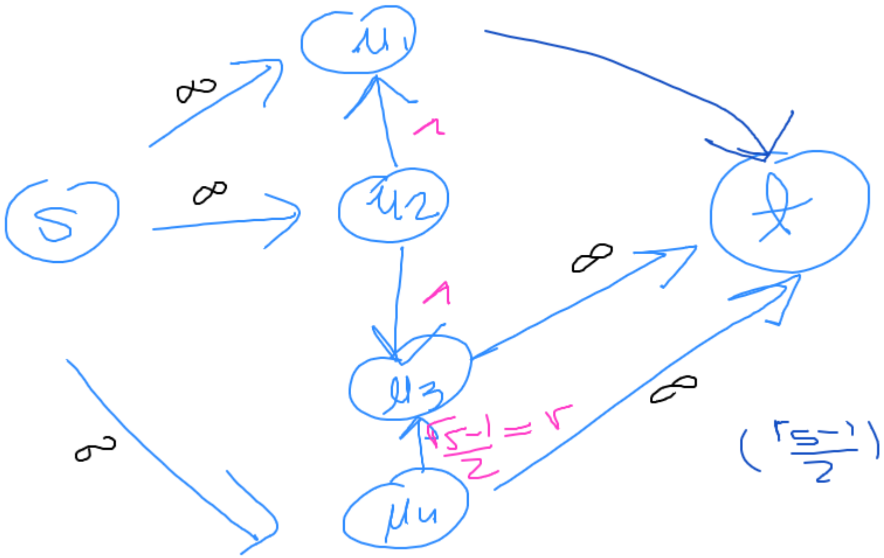
// $O(f^* \cdot n)$

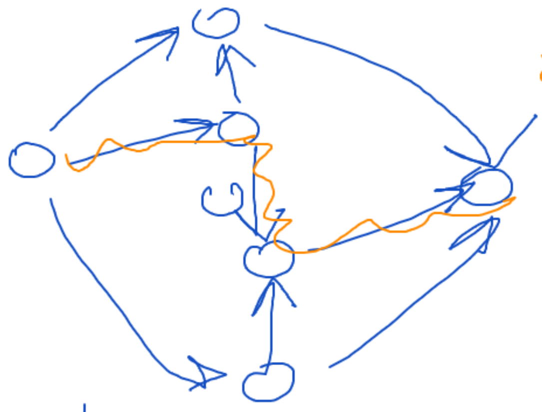
si es racional $\rightarrow O(f^* n \cdot D)$

si no es racional?

Egm

$r = \frac{\sqrt{5}-1}{2} < 1$





$\varepsilon' = 1$

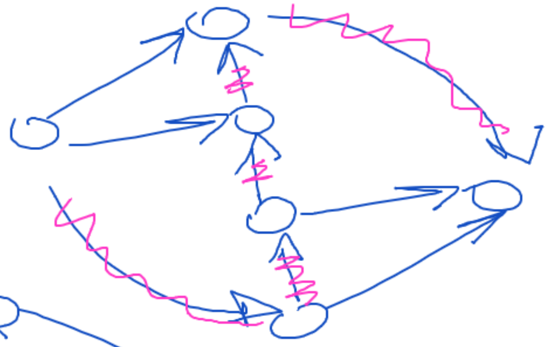
$f = 1$

$\left(\frac{\sqrt{5}-1}{2}\right)^2 = \frac{6-2\sqrt{5}}{4} = \frac{3-\sqrt{5}}{2}$

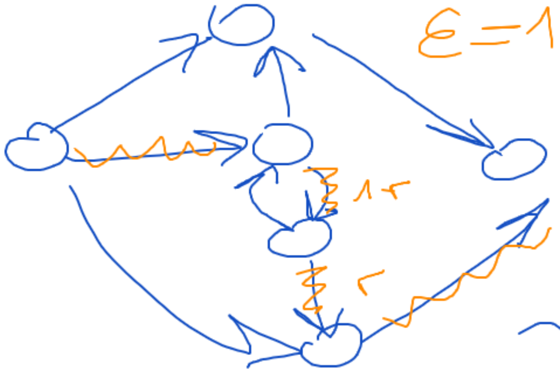
$\frac{\sqrt{5}-1}{2} > \frac{1}{2}$

$1-r < r$

$\varepsilon = r$

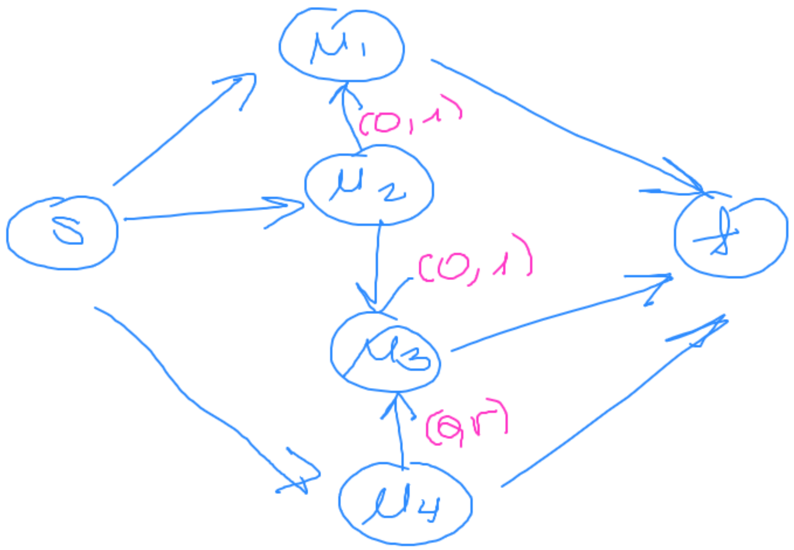


$\varepsilon = 1 - r = r^2$



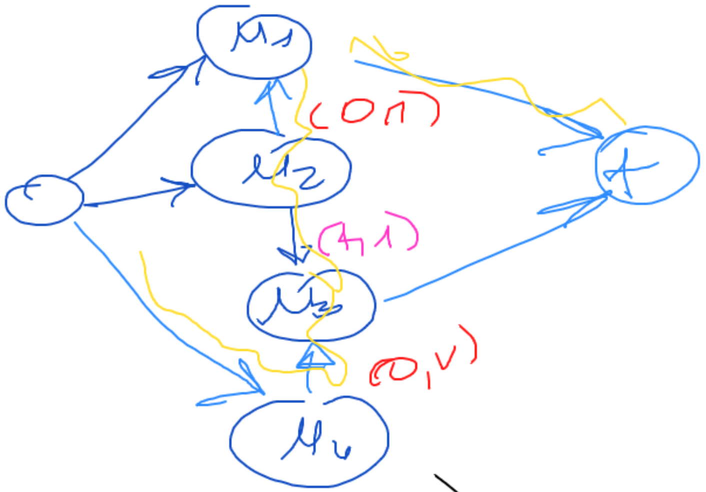
$1-r < r$

$r > \frac{1}{2}$



$\varepsilon = 1$

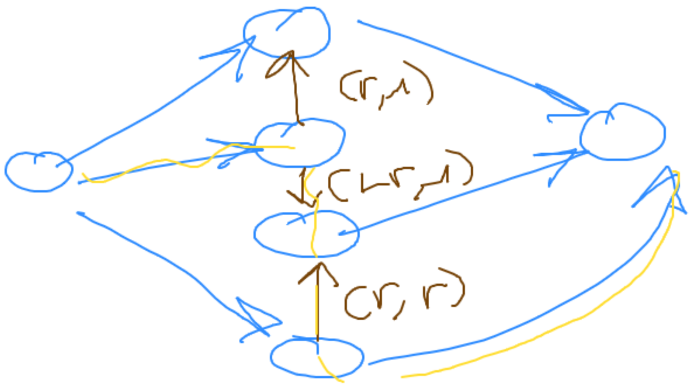
$s u_2 u_3$

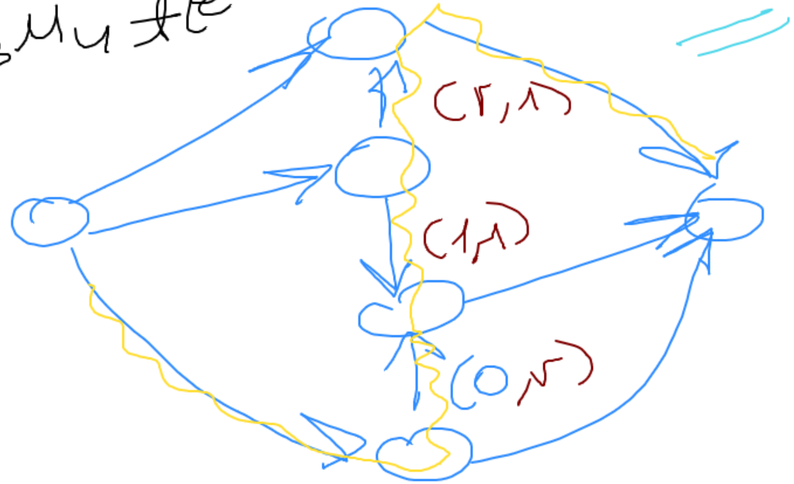


$\varepsilon = 1-r$

$s u_4 u_3 u_2 u_1 t$ $(\varepsilon = r)$

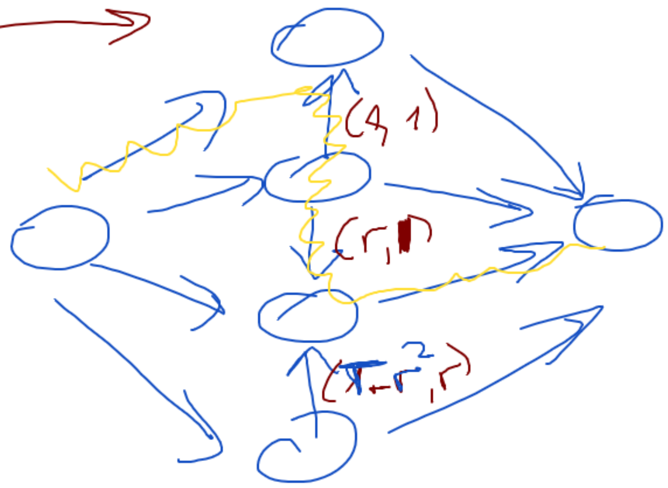
$s u_2 u_3 u_4 t$ $(\varepsilon = r)$



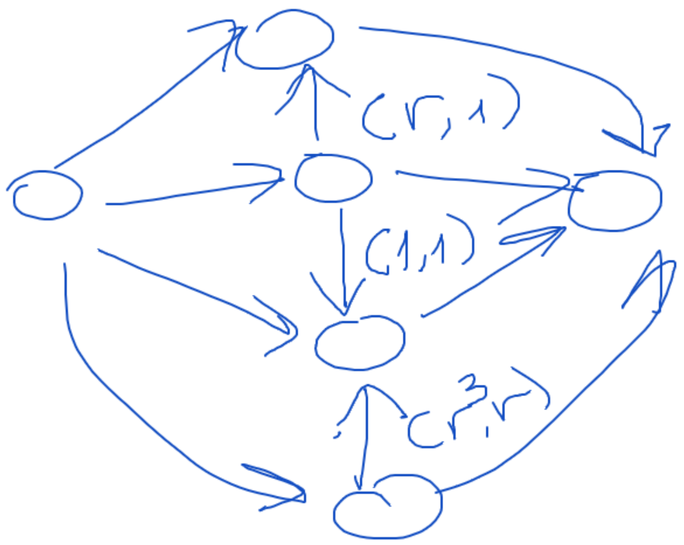


$\varepsilon = r^2$

$\longrightarrow$



$\Rightarrow$



r^3

$\longrightarrow$

$$r - r^2 = r^3$$